JILIN TONGYU TONGFA WIND POWER PROJECT

Document Prepared by Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Jilin Tongyu Tongfa Wind Power Project (hereafter referred as the project) is a grid connected renewable energy project located in Tongyu County, Baicheng, Jilin Province, Northeast of China. The objective of the project is to generate electricity using wind power generation technology and to sell into China Northeast Power Grid. The project achieves CO_2 emission reduction by replacing electricity generated by fossil fuel fired power plant.

Totally 118 wind turbines with a nominal capacity of 850 KW have been installed, providing a total capacity of 100.3MW. With the annual electricity delivered to the grid of 227,200MWh, the project is estimated to deliver 249,920 tCO_2 emission reduction annually.

The project began to construct on 15-April-2007. The first set of wind turbines started official operation on 06-December-2007[1], and the project started full operation on 23-November-2008.

During this monitoring period, the operation condition of all the wind turbines, the monitoring system and the operation environment haven't been changed from the previous monitoring periods, which are also consistent with the description of the registered PD.

This is the second monitoring period from 25-April-2012 to 05-December-2017 under VCS program for the project. The total GHG emission reductions generated in this monitoring period are 716,004 tCO_2e.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type: Grid connected wind power project.

The project is not a grouped project.

1.3 Project Proponent

Organization name	Jilin Longyuan Wind Power Co., Ltd.
Contact person	Mr. XU Xiangbo

[1] The monitoring reports of previous CDM verifications stated that the project the first wind turbine started operation on 20-November-2007. However, the first wind turbine actually started commissioning on 20-November-2007 and was put into official operation on 06-December-2007. Thus, start date of VCS crediting period of the project is confirmed to be 06-December-2007 on which the project starting official operation.

Title	General Manager
Address	Room 1105, Guangming Building, No. 336 Xi'an Road, Changchun, Jilin, China
Telephone	+8643180583962
Email	12106274@chnenergy.com.cn

1.4 Other Entities Involved in the Project

Organization name	**Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.**
Role in the Project	Consultant
Contact person	Mrs. YU Shanshan
Title	General Manager
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1.5 Project Start Date

06-December-2007 (The date of first set of wind turbines started official operation).

1.6 Project Crediting Period

The first 10-year crediting period of the project under VCS is from 06-December-2007 to 05-December-2017 (both days included).

1.7 Project Location

The project is located in Tongyu County, Baicheng, Jilin Province of China, its geographical coordinates are north latitude 44° 13' N ~ 45° 16' N and east longitude 122° 02' E ~ 123° 30'E, and its altitude is 170-200 m. The detailed location of the project is shown in Figure 1.

Figure 1 The Site of the project

1.8 Title and Reference of Methodology

Applied methodology: ACM0002 (Version 6): Consolidated monitoring methodology for zero-emissions grid-connected electricity generation from renewable sources.

Applied tools: Tool for the Demonstration and Assessment of Additionality (version 04).

The applied methodology and tool can be found from the following link:

https://cdm.unfccc.int/methodologies/PAmethodologies/approved

1.9 Participation under other GHG Programs

The project is a registered CDM project with reference No. 2068, the detailed information can be found at https://cdm.unfccc.int/Projects/DB/DNV-CUK1218468288.41/view

Start Date	End Date	GHG Emission Reduction (tCO_2e)	Credit	Program
06-December-2007	25-January-2009	130,457	VCU	VCS
26-January-2009	24-July-2009	88,062	CER	CDM
25-July-2009	24-June-2010	148,716	CER	CDM

25-June-2010	24-June-2011	171,651	CER	CDM
25-June-2011	24-April-2012	84,355	CER	CDM

This is the second monitoring period under VCS program from 25-April-2012 to 05-December-2017 for the project. All emission reductions during this monitoring period have not and will not seek other kind of GHG programs issuance. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO_2e/year, not including renewable project[2].

The project proponent Jilin Longyuan Wind Power Co., Ltd., as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS. Therefore, the project is not involved in an emission trading program or any other mechanism that includes GHG allowance trading. Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China.

Furthermore, a statement on no double counting will be submitted to Verra to confirm the credits during this monitoring period has not been counted and will not be counted under emission trading programs and other binding limits.

Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The project activity is a renewable wind power project located in Jilin province, Northeast China.

[2] http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

The project utilizes wind resources to generate and supply 650,915.700 MWh renewable electricity to the power grid annually, which contributes to SDG 7.

The project provides 97 long-term job opportunities for local residents, which has a positive effect on the local economy and contributes to SDG 8.

The renewable electricity generated by the project replaces electricity generation by fossil-fuel power plants of the power grid, which reduces 716,004 tCO_2e GHG emission annually and contributes to SDG 13.

Please refer to section 4 and section 5 of this monitoring report for the evidence of the project's contribution to SD 7 and SDG 13. And the evidence for SDG 8 is submitted as a separate document to Verra due to confidentiality requirement of the project proponent.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	650,915.700 MWh of electricity from renewable sources has been supplied to the power grid during the reporting period	An accumulated 650,915.700 MWh of electricity from renewable sources has been supplied to the power grid by end of the reporting period
2)	8.5	Numbers of job opportunities provided by implementation of the project	Implemented activities to increase	97 people have been employed by the project as long-term employee	97 people have been employed by the project as long-term employee by end of the reporting period
3)	13.0	Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	The project has achieved a total GHG emission reduction of 716,004 tCO_2e during the reporting period.	An accumulated 716,004 tCO_2e of GHG emission reduction have been achieved by the project by end of the reporting period

2 SAFEGUARDS

2.1 No Net Harm

Environmental Impact

The Environmental Impact Assessment (EIA) of the project has been conducted and approved by local government officials. The EIA report has assessed every possible aspect of environmental impacts of the project and recommended corresponding measures, where applicable.

The EIA shows that the air quality of the project site is relatively well, which meets the Class I of China Ambient Air Quality Standard (GB 3095-1996); the environmental noise meets Class II of China Environmental Noise Standard in Urban Area (GB 3096-93); and the quality of groundwater meets the Class III of China Quality Standard for Ground Water (GB/T 14848-93), which can be used as drinking water resource. The EIA shows that the environmental impact will occur during the construction period and operation period.

The construction period: The waste water released during the construction period will be mainly living waste water. A leakage proof cesspool will be built in the living area for construction staff. The waste water will be treated in the cesspool and then be reused as irrigation water, and will thus have little impact on the surface water. The solid living waste will be transported to landfills. Dumping waste in the proposed project site is forbidden to minimize environmental impacts. The impacts on the sound environment and air quality are short-term during the construction period. After the completion of the construction, the impacts will disappear naturally. The occupation of ground will destroy some surface vegetation during the construction period, but the vegetation destroyed by temporary ground occupation will be recovered after the completion of the construction. The impacts on local ecology system will then be minimized.

The operation period: There will be no air pollutant emissions during the operation period of the proposed project. There will be a little amount of living waste water released during the operation period of the proposed project, which will be treated in the leakage proof cesspool and then be reused in farming irrigation. Therefore, the waste water will have little impact on the surface water. The noise source during the operation period of the proposed project will be mainly from the running of the wind power unit, which will have no impacts on the area 500m away. Since the site of the proposed project is far from the living areas, there will be no noise impacts on local environment. The electromagnetism of wind power project operation will be relatively low, and will have little impact on the neighbourhood environment. The operation of the proposed project will have slight impacts on birds.

In conclusion, being as a typical type of clean renewable energy, the proposed project has no significant impacts on local environment and will greatly contribute to achievement of sustainable development objective and promote local environmental protection, thus, no net harm on local environment and social community has been detected for the project.

Socio-economic impact

During the construction and operation of this project, the project provides employment opportunities for local residents. And regularly provide vocational training for employees to help employees set up career plans. Therefore, the project has played an important role in the local social and economic development. Please refer to section 1.11 of this report for details.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

On July 26th, 2006, under the support of the local municipal government, the project owner successfully held a stakeholder meeting in Tongyu County. Totally 8 stakeholder representatives participated the meeting, respectively from the local Government, local Environmental Protection Bureau, local Development and Reform Bureau, local Agricultural Power Bureau, and Tongfa Village in Tongyu County.

Every stakeholder representative expressed the comments for the proposed project. No opposite comment was received. The summary of the comments is as follows: Comments from the local government: The proposed project has been approved by the Development and Reform Commission and Environmental Protection Administration of Jilin Province, which shows that the construction and operation of the proposed project will have little impacts on the local environment. The proposed project is one of the largest investment projects in local area. The local government highly supports the proposed project, and expects the increase of local financial incoming and new employment opportunity through the implementation of the proposed project. Comments from the local Agricultural Power Bureau: Wind Power is renewable energy and is helpful in diversifying power mix of local power grid. Furthermore, wind power takes up a small share in the local power grid and will not have much impact on the local grid. Comments from villager representatives: The proposed project site is located at a badlands. There are no residents and only a few croplands around the proposed project. Therefore, there is no issue on noise disturbance and residents movement. Moreover, the project owner has made compensation for the land occupied by the proposed project. The local residents also benefit from the employment opportunities for construction and operation of the proposed project.

LSC during the operation period

Communications with Local stakeholders have been carried out at periodic intervals. Key implementation schedules and changes of the project have been communicated to the local authority, the neighbourhood committee, and local residents. The comments and suggestions from residents were collected by the local authority, if any.

The project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in June 2012, July 2014 and May 2016 during this monitoring period. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations. There have been no negative comments received for the project so far.

The following people are considered as the stakeholders of this project during the implementation stage.

- Residents of the nearby village.
- Staff of the project.
- Relevant administrative staff of the local government.

The project owner designed a questionnaire to collect the comments of relevant stakeholders and the major question issues are shown as below:

1. Are you satisfied with the current environmental quality of the project location?
2. What do you think are the main environmental problems in the current location of the project?
3. In your opinion, which environmental protection measures need to be strengthened during the operation period of this project?
4. What impact do you think this project activity may have on local employment and residents' income?
5. Impact on the local economic development.
6. Attitude on implementation of the project.

30 questionnaires have been handed in total and all of them have been received successfully from the stakeholder representatives, respectively from the local government, and surrounding villages. The representatives covered different gender, different ages, different occupations and different education levels. The detail information is shown as below:

Item		Percentage		
		June-2012	July-2014	May-2016
Amount of stakeholders surveyed	Male	50%	40%	60%
	Female	50%	60%	40%
Age	<25	6.67%	0%	3.33%
	25-45	16.67%	16.67%	16.67%
	45-55	66.66%	66.66%	16.67%
	>55	10%	16.67%	63.33%
Education	Junior high school or below	50%	40%	60%

	Senior high school	50%	50%	40%
	College or above	0%	10%	0%
Occupation	Worker	26.67%	16.67%	26.67%
	Farmer	26.67%	63.33%	26.67%
	Management personnel	6.66%	3.33%	6.66%
	Unspecified	40%	16.67%	40%

All local stakeholders are supportive to the implementation of the project, believing that the Project help improve the life of local people and promote local economic development without adverse environmental impact.

However, during the operation, some main issues were raised by local stakeholder and the developer should pay attention and solve these issues which are listed as following:

1. Some measurements should be taken to control the environmental pollution;

2. Project developer should ensure the security of the power station during the operation period;

For the first issue, as the project merely caused environmental pollution during the operation period, this issue can be easily controlled; for the second issue, during this monitoring period, all the workers received the safety training regularly.

Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations. There are no negative comments received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

Totally 118 Gamesa58-850KW wind turbines with a nominal capacity of 850 kW were installed, providing a total capacity of 100.3MW. The main technical specifications are as follows:

Diameter	58m
Swept Area	2124m^2
Rotational Speed Operation Interval	14.6-30.8rpm (55/65m towers)
Sense of Rotation	Clockwise (front view)
Rotor Orientation	Upwind
Tilt angle	6°
Blade coning	3°
Number of blades	3
Aero-dynamic brake	Full feathering

Each turbine has a 0.69/35kV transformer, from which a 35kV line is link into the on-site 220kV switchgear at the substations established in the project site. By the 220kV line, the electricity generated by the project is delivered to the power grid.

The auxiliary electric system of wind farm includes onsite control, protection, measure, signalling and surveillance in central control room of wind farm. The targets to be surveilled include 118 wind turbines and transformers. The information of the wind farm is dispatched by Province level dispatch centre, and wind turbines could be measured and signalled remotely from Changchun City by Internet. The wind turbines and transmission facility could be monitored and controlled either by onsite central control room or remotely through Internet.

During this monitoring period, the project was implemented in accordance with the registered VCS PD (i.e. registered CDM PDD version 4.1 dated 20/01/2009). The Project has operated without any accidental or emergency events that might impact the GHG emission reductions or removals and monitoring.

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation in this monitoring period

3.2.2 Project Description Deviations

There is a deviation for the crediting period. The project is registered under VCS Version 2007.1 and completed validation before 19--March-2020. thus, it remains eligible to apply the crediting period requirements under VCS version 3 which shall be a maximum of ten years and may be renewed at most twice so the 1st crediting period of the project should be updated from “06-December-2007 to 25-January-2009” to “06-December-2007 to 05-December-2017”. The project did not complete the validation of the crediting period update within two years of the end of the first crediting period, therefore the VCS total crediting period ends on 05-December -2017, the end of the first crediting period.

The deviation does not impact the applicability of the methodology, additionality and the appropriateness of the baseline scenario.

3.3 Grouped Projects

Not applicable.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF_y
Data unit	tCO_2/MWh
Description	Baseline emission factor
Source of data	Registered PD (i.e. registered CDM PDD version 4.1 dated 20/01/2009)
Value applied	1.1000
Justification of choice of data or description of measurement methods and procedures applied	Ex-ante determined in the registered PD and fixed during the 1st crediting period.
Purpose of Data	Calculation of baseline emissions
Comments	Determined ex-ante and fixed for the 1st crediting period

4.2 Data and Parameters Monitored

<table>
<tr><th>Data / Parameter</th><th>EG_y</th></tr>
<tr><td>Data unit</td><td>MWh</td></tr>
<tr><td>Description</td><td>Electricity delivered to the grid by the project</td></tr>
<tr><td>Source of data</td><td>Measured by electricity meters: M1</td></tr>
<tr><td>Description of measurement methods and procedures to be applied</td><td>Two meters are installed, the key meter M1 and the backup meter M2. The two meters (accuracy degree is 0.2S, bidirectional) are installed at the outlet of the on-site 220KV transformer substation.

Net electricity supplied by the project to the grid is the difference between the electricity exported to and the electricity imported from the grid.</td></tr>
<tr><td>Frequency of monitoring/recording</td><td>Continuously measured daily read and monthly recorded</td></tr>
<tr><td>Value monitored</td><td>650,915.700</td></tr>
<tr><td>Monitoring equipment</td><td><table>
<tr><td>Monitoring equipment</td><td>M1（key meter）</td><td>M2 (backup meter)</td></tr>
<tr><td>Type</td><td>ZMD402CT44.0257 S2</td><td>ZMD402CT44.0257 S2</td></tr>
<tr><td>Serial No.</td><td>86688100</td><td>86688095</td></tr>
</table></td></tr>
</table>

<table>
<tr><td></td><td>
<table>
<tr><td>Accuracy</td><td>0.2S</td><td>0.2S</td></tr>
<tr><td>Calibration frequency</td><td>Annually</td><td>Annually</td></tr>
<tr><td>Calibration date</td><td>02-March-2012
28-February-2013
27-February-2014
24-February-2015
22-February-2016
20-February-2017</td><td>02-March-2012
28-February-2013
27-February-2014
24-February-2015
22-February-2016
20-February-2017</td></tr>
<tr><td>Calibration validity</td><td>01-March-2013
27-February-2014
26-February-2015
23-February-2016
21-February-2017
19-February-2018</td><td>01-March-2013
27-February-2014
26-February-2015
23-February-2016
21-February-2017
19-February-2018</td></tr>
</table>
</td></tr>
<tr><td>QA/QC procedures to be applied</td><td>Electricity meters (accuracy degree is 0.2S, bidirectional) have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448). Meter readings crosschecked with electricity sales receipts.

Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup. The project owner is responsible for the measurements.</td></tr>
<tr><td>Purpose of the data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Calculation method</td><td>The electricity exported to the grid minus the electricity imported from the grid.</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

4.3 Monitoring Plan

1. Organizational structure

The Monitoring Plan clearly states the persons and the responsibilities of the QC team of the project; they are responsible for monitoring the electricity generation of the project. The assigned staff is in accordance with the registered PD.

The organizational structure and responsibilities of the personnel is shown as follows:

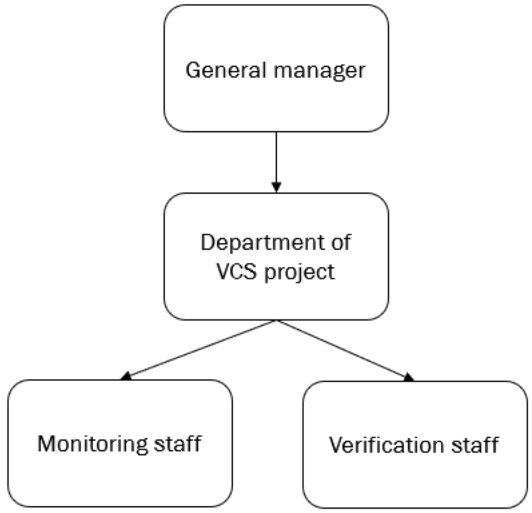

Figure 2 Organizational structure

Responsibility

General manager:

1. Appoint VCS monitoring persons and supervise the monitoring.

2. Take the overall responsibility of the VCS monitoring.

3. Internal auditing

CDM manager:

1. Cooperate with the VVB and the consultancy company to complete validation and verification;

2. Collect the project completion report, power purchase agreement, calibration organization qualification and meter calibration report;

3. Communicate and coordinate with the meter calibration organization when the meter is out of order

4. Organize the training of the relative persons.

5. Check daily reading report and monthly reading report.

Verification person

1. Collect the copy of sales receipts for each month;

2. Check the records provided by the grid monthly with the records from the wind farm. If any data errors are detected, inform the monitoring responsible person and correct the data immediately.

Monitoring person

1. Carry out operational rules and regulations made for the wind farm;

2. Organize and dispatch workers

3. Ensure the equipment run normally. Report the abnormal to the CDM manager.

4. Be responsible for the monthly electricity statistics. And send the result to the auditing responsible person.

5. Record the readings of the meters

6. Check the records with the readings recorded by the wind turbine system. If any data errors are detected, correct the data immediately

2. Monitoring equipment

Two bidirectional electronic meters are installed for the project, as shown in the figure below. They are installed at the outlet of the 220KV transformer substation of the power grid (with an accuracy of 0.2S) to measure electricity delivered to the grid by the project which is difference of the electricity exported to the grid by the project and the electricity imported from the grid.

For electricity exported to and imported from the grid, the key meter was used in normal situation. Readings of backup meter is used in case of the key meter is out of work. No error was occurred during this monitoring period. The designated person of the project recorded the reading of the meters at 0:00 daily, and then the electricity exported to and imported from the grid by the project were aggregated at 24:00 o'clock on the 24th of every month. After reading the key meter and confirming the reading with the project owner, the grid issues the sales receipts based on the checked electricity amount. The sales receipts are used for double checking purpose to ensure consistency.

All the data and relevant documents are archived for 2 years following the end of the crediting period by means of electronic and paper backup

The installation of electricity meters is shown in the figure below:

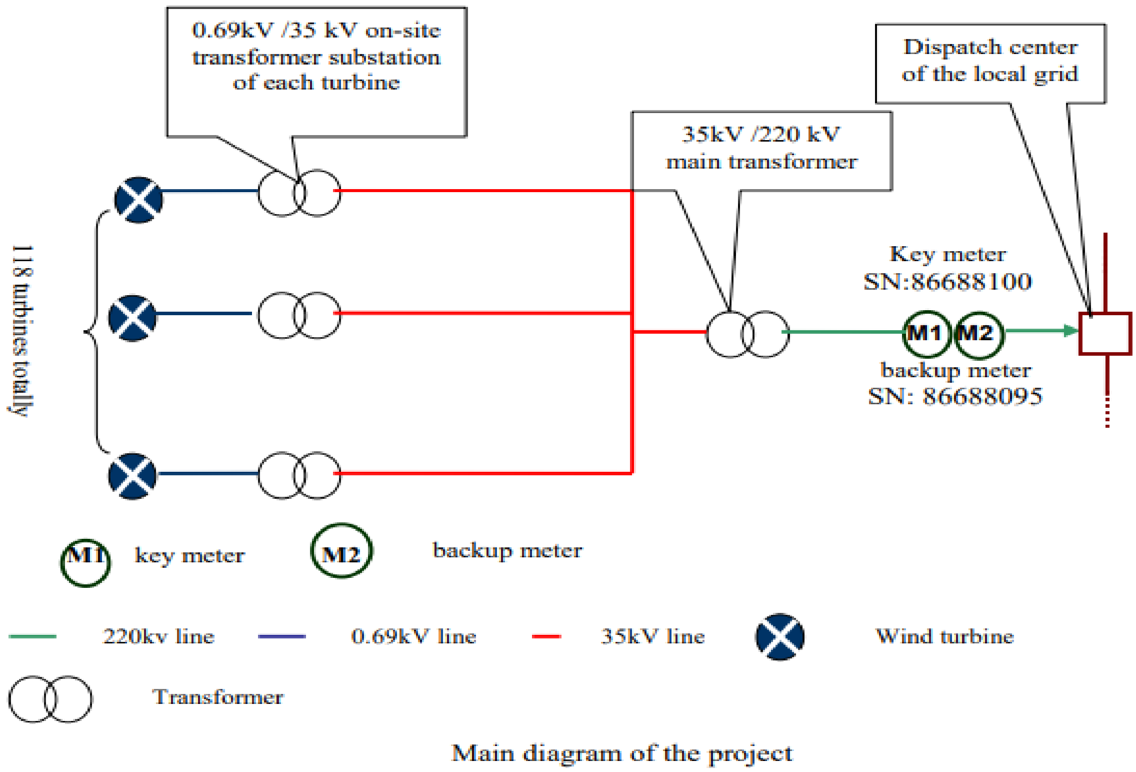

Figure 3: Installation of the metering system

3. Monitoring Practice and Handling of Emergency

The key meter is used to measure the electricity exported to the grid by the project, and as a bidirectional meter, also the electricity imported from the grid by the project. In case of emergency of the key meter, e.g., when it does not work, the readings of the backup meter is used as a reference to determine the electricity exported to and imported from the grid by the project, and thus the electricity delivered to the grid by the project.

4. Data Management System

In order to facilitate verification, all data relevant to monitoring are carefully kept and stored by the project owner. The following table below outlines the key documents relevant to monitoring and verification of the emission reductions of the project.

No.	Document	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation	PO

F-2	The monitoring report on the electricity delivered to the power grid	PO

5. Training

The monitoring staff has all received sufficient training in terms of monitoring and verification. They have received general training on wind power project operation organized by the project owner, including reading and calibration of meters, recording of meter readings, and reporting of readings. They all have the professional qualifications of operation and maintenance in electricity power industry. Furthermore, they have received VCS training organized by the project owner.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

As per the registered PD, the baseline emission of the project is calculated as below:

$$BE_y = EG_y \times EF_y \quad (1)$$

Where,

BE_y	Baseline emissions in year y (t CO_2e)
EG_y	Electricity delivered to the grid by the project (MWh)
EF_y	Baseline emission factor in the year y (tCO_2e/MWh)

The monitored EG_y is reported in the table below.

Table 1 Monitored monthly value of EG_y[3]

Year	Month	Electricity delivered to the grid by the project			Electricity Imported from the grid by the project			
		Monthly recording data from M1(MWh)	Sales Receipt (MWh)	Conservative quantity	Monthly recording data from M1 (MWh)	Sales Receipt (MWh)	Conservativ e quantity	EG_y
		A	B	C=min(A,B)	D	E	F=max(D,F)	G=C-F
2012	25-April~24-May	9,820.000	9,820.000	9,820.000	42.530	42.530	42.530	9,777.470
	25-May~24-June	5,000.000	5,000.000	5,000.000	202.600	202.600	202.600	4,797.400
	25-June~24-July	7,640.000	7,640.000	7,640.000	96.043	96.043	96.043	7,543.957
	25-July-24-August	8,590.000	8,590.000	8,590.000	102.960	102.960	102.960	8,487.040
	25-August~24-September	5,330.000	5,330.000	5,330.000	109.190	109.190	109.190	5,220.810
	25-September~24-October	11,240.000	11,240.000	11,240.000	75.425	75.425	75.425	11,164.575
	25-October~24-November	12,600.000	12,600.000	12,600.000	71.676	71.676	71.676	12,528.324
	25-November~24-December	11,930.000	11,930.000	11,930.000	175.481	175.481	175.481	11,754.519

[3] The electricity data for the period from 25th December to 31st December, 2012-2016, 1st January to 24th January, 2013-2017 and 25th November to 5thDecember, 2017 were obtained from the project company's electricity records report, which was approved by the local grid company.

	25-December~31-December	1,939.677	8,590.000	1,939.677	69.729	69.729	69.729	1,869.948
	Subtotal	**74,089.677**	**80,740.000**	**74,089.677**	**945.634**	**945.634**	**945.634**	**73,144.043**
2013	01-January~24-January	6,650.323	8,590.000	6,650.323	239.072	239.072	239.072	6,411.251
	25-January~24-February	11,220.000	11,220.000	11,220.000	83.054	83.054	83.054	11,136.946
	25-February~24-March	17,630.000	17,630.000	17,630.000	45.725	45.725	45.725	17,584.275
	25-March~24-April	20,420.000	20,420.000	20,420.000	43.639	43.639	43.639	20,376.361
	25-April~24-May	19,180.000	19,180.000	19,180.000	303.860	303.860	303.860	18,876.140
	25-May~24-June	14,150.000	14,150.000	14,150.000	69.406	69.406	69.406	14,080.594
	25-June~24-July	8,470.000	8,470.000	8,470.000	120.622	120.622	120.622	8,349.378
	25-July-24-August	6,760.000	6,760.000	6,760.000	85.219	85.219	85.219	6,674.781
	25-August~24-September	11,820.000	11,820.000	11,820.000	89.998	89.998	89.998	11,730.002
	25-September~24-October	11,970.000	11,970.000	11,970.000	80.362	80.362	80.362	11,889.638
	25-October~24-November	11,480.000	11,480.000	11,480.000	48.418	48.418	48.418	11,431.582
	25-November~24-December	9,970.000	9,970.000	9,970.000	130.680	130.680	130.680	9,839.320
	25-December~31-December	1,684.516	7,460.000	1,684.516	32.167	32.167	32.167	1,652.349
	Subtotal	**151,404.839**	**159,120.000**	**151,404.839**	**1,372.222**	**1,372.222**	**1,372.222**	**150,032.617**

2014	01-January~24-January	5,775.484	7,460.000	5,775.484	110.287	110.287	110.287	5,665.197
	25-January~24-February	5,150.000	5,150.000	5,150.000	169.356	169.356	169.356	4,980.644
	25-February~24-March	11,020.000	11,020.000	11,020.000	110.167	110.167	110.167	10,909.833
	25-March~24-April	12,520.000	12,520.000	12,520.000	53.539	53.539	53.539	12,466.461
	25-April~24-May	10,560.000	10,560.000	10,560.000	90.922	90.922	90.922	10,469.078
	25-May~24-June	9,200.000	9,200.000	9,200.000	98.076	98.076	98.076	9,101.924
	25-June~24-July	10,750.000	10,750.000	10,750.000	93.588	93.588	93.588	10,656.412
	25-July-24-August	4,070.000	4,070.000	4,070.000	162.545	162.545	162.545	3,907.455
	25-August~24-September	7,850.000	7,850.000	7,850.000	92.083	92.083	92.083	7,757.917
	25-September~24-October	8,740.000	8,740.000	8,740.000	95.621	95.621	95.621	8,644.379
	25-October~24-November	10,750.000	10,750.000	10,750.000	73.946	73.946	73.946	10,676.054
	25-November~24-December	12,050.000	12,050.000	12,050.000	136.541	136.541	136.541	11,913.459
	25-December~31-December	1,772.581	7,850.000	1,772.581	35.428	35.428	35.428	1,737.153
	Subtotal	**110,208.065**	**117,970.000**	**110,208.065**	**1,322.099**	**1,322.099**	**1,322.099**	**108,885.966**
2015	01-January~24-January	6,077.419	7,850.000	6,077.419	121.467	121.467	121.467	5,955.952
	25-January~24-February	7,100.000	7,100.000	7,100.000	48.286	48.286	48.286	7,051.714

	25-February~24-March	9,600.000	9,600.000	9,600.000	48.286	48.286	48.286	9,551.714
	25-March~24-April	17,790.000	17,790.000	17,790.000	30.809	30.809	30.809	17,759.191
	25-April~24-May	13,030.000	13,030.000	13,030.000	57.684	57.684	57.684	12,972.316
	25-May~24-June	8,190.000	8,190.000	8,190.000	75.029	75.029	75.029	8,114.971
	25-June~24-July	3,860.000	3,860.000	3,860.000	109.560	109.560	109.560	3,750.440
	25-July-24-August	3,770.000	3,770.000	3,770.000	121.836	121.836	121.836	3,648.164
	25-August~24-September	5,440.000	5,440.000	5,440.000	83.450	83.450	83.450	5,356.550
	25-September~24-October	12,680.000	12,680.000	12,680.000	43.560	43.560	43.560	12,636.440
	25-October~24-November	11,760.000	11,760.000	11,760.000	114.814	114.814	114.814	11,645.186
	25-November~24-December	6,440.000	6,440.000	6,440.000	186.886	186.886	186.886	6,253.114
	25-December~31-December	1,833.548	8,120.000	1,833.548	27.851	27.851	27.851	1,805.697
	Subtotal	**107,570.968**	**115,630.000**	**107,570.968**	**1,069.518**	**1,069.518**	**1,069.518**	**106,501.449**
2016	01-January~24-January	6,286.452	8,120.000	6,286.452	95.490	95.490	95.490	6,190.962
	25-January~24-February	6,850.000	6,850.000	6,850.000	101.402	101.402	101.402	6,748.598
	25-February~24-March	9,290.000	9,290.000	9,290.000	49.421	49.421	49.421	9,240.579
	25-March~24-April	17,660.000	17,660.000	17,660.000	44.748	44.748	44.748	17,615.252

	25-April~24-May	12,930.000	12,930.000	12,930.000	48.576	48.576	48.576	12,881.424
	25-May~24-June	9,520.000	9,520.000	9,520.000	87.569	87.569	87.569	9,432.431
	25-June~24-July	7,770.000	7,770.000	7,770.000	88.123	88.123	88.123	7,681.877
	25-July-24-August	3,730.000	3,730.000	3,730.000	83.292	83.292	83.292	3,646.708
	25-August~24-September	5,940.000	5,940.000	5,940.000	128.093	128.093	128.093	5,811.907
	25-September~24-October	7,890.000	7,890.000	7,890.000	83.741	83.741	83.741	7,806.259
	25-October~24-November	9,950.000	9,950.000	9,950.000	104.016	104.016	104.016	9,845.984
	25-November~24-December	10,190.000	10,190.000	10,190.000	102.010	102.010	102.010	10,087.990
	25-December~31-December	1,406.774	6,230.000	1,406.774	27.541	27.541	27.541	1,379.233
	Subtotal	**109,413.226**	**116,070.000**	**109,413.226**	**1,044.022**	**1,044.022**	**1,044.022**	**108,369.204**
2017	01-January~24-January	4,823.226	6,230.000	4,823.226	94.427	94.427	94.427	4,728.798
	25-January~24-February	5,990.000	5,990.000	5,990.000	40.471	40.471	40.471	5,949.529
	25-February~24-March	12,470.000	12,470.000	12,470.000	96.703	96.703	96.703	12,373.297
	25-March~24-April	13,040.000	13,040.000	13,040.000	40.207	40.207	40.207	12,999.793
	25-April~24-May	16,130.000	16,130.000	16,130.000	36.643	36.643	36.643	16,093.357
	25-May~24-June	9,260.000	9,260.000	9,260.000	76.507	76.507	76.507	9,183.493
	25-June~24-July	6,100.000	6,100.000	6,100.000	176.220	176.220	176.220	5,923.780

	25-July-24-August	3,040.000	3,040.000	3,040.000	111.487	111.487	111.487	2,928.513
	25-August~24-September	9,410.000	9,410.000	9,410.000	53.566	53.566	53.566	9,356.434
	25-September~24-October	9,560.000	9,560.000	9,560.000	78.778	78.778	78.778	9,481.222
	25-October~24-November	12,540.000	12,540.000	12,540.000	38.993	38.993	38.993	12,501.007
	25-November~05-December	2,577.667	7,030.000	2,577.667	114.470	114.470	114.470	2,463.197
	Subtotal	**104,940.892**	**110,800.000**	**104,940.892**	**958.472**	**958.472**	**958.472**	**103,982.421**
	Total	**657,627.667**	**700,330.000**	**657,627.667**	**6,711.967**	**6,711.967**	**6,711.967**	**650,915.700**

The baseline emissions of the project are calculated as follows:

Table 2 Calculation of Baseline emissions

Year	EG_y (MWh)	EF_y (tCO_2/MWh)	BE_y (tCO_2)
25-April-2012 to 31-December-2012	73,144.043	1.1000	80,458
01-January-2013 to 31-December-2013	150,032.617	1.1000	165,035
01-January-2014 to 31-December-2014	108,885.966	1.1000	119,774
01-January-2015 to 31-December-2015	106,501.449	1.1000	117,151
01-January-2016 to 31-December-2016	108,369.204	1.1000	119,206
01-January-2017 to 05-December-2017	103,982.421	1.1000	114,380
Total	650,915.700		716,004

5.2 Project Emissions

As per the applied methodology, the project emission is zero.

5.3 Leakage

As per the applied methodology, leakage of the project is not considered.

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
25-April-2012 to 31-December-2012	80,458	0	0	80,458
01-January-2013 to 31-December-2013	165,035	0	0	165,035
01-January-2014 to 31-December-2014	119,774	0	0	119,774
01-January-2015 to 31-December-2015	117,151	0	0	117,151

01-January-2016 to 31-December-2016	119,206	0	0	119,206
01-January-2017 to 05-December-2017	114,380	0	0	114,380
Total	716,004	0	0	716,004

Comparison of actual emission reduction and the estimation in the registered PD.

Years	Emission reduction achieved (tCO_2)	Estimated ER in registered PD (tCO_2)	Difference
25-April-2012 to 05-December-2017	716,004	1,404,345[4]	-49.02%

The actual emission reduction achieved during this monitoring period is 49.02[5]% lower than the estimation for the same period in the registered PD due to the insufficient capacity of the grid and wind fluctuations. Refer to ER calculation spreadsheet for detailed calculation process.

[4] 1,404,345 tCO_2 = 249,920 tCO_2 * 2051days/365days

[5] 49.02% = (716,004 tCO_2/1,404,345 tCO_2 – 1) * 100%

APPENDIX1: <SUPPORTING EVIDENCE>

Evidence for SDG 7 and SDG 13: From project operation date onwards, the net electricity supplied to the grid and corresponding emission reduction in every monitoring period have been listed in below table. Please refer to following links for details.

https://cdm.unfccc.int/Projects/DB/DNV-CUK1218468288.41/view

https://registry.verra.org/app/projectDetail/VCS/276

Start Date	End Date	EG_y (MWh)	ER (tCO_2e)
06-December-2007	25-January-2009	118,597.2727	130,457
26-January-2009	24-July-2009	80,056.5744	88,062
25-July-2009	24-June-2010	135,197.1456	148,716
25-June-2010	24-June-2011	156,046.5720	171,651
25-June-2011	24-April-2012	76,687.2480	84,355
25-April-2012	05-December-2017	650,915.700	716,004

Evidence for SDG 8: The employee list of the project has been submitted as a separate document to VVB due to confidentiality requirement of the project proponent.